

International Journal of Engineering Sciences & Research Technology

(A Peer Reviewed Online Journal)
Impact Factor: 5.164



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**INTERNATIONAL JOURNAL OF ENGINEERING SCIENCES & RESEARCH
TECHNOLOGY****ENHANCING FANET RELIABILITY AND CONGESTION CONTROL WITH AN
IMMACULATE AND DEXTEROUS EFCCC ROUTING PROTOCOL****P. Sunil¹, Dr. V. Raghunatha Reddy²**¹Research Scholar, Dept. of Computer Science & Technology, Sri Krishnadevaraya University, AP, India.²Professor, Dept. of Computer Science & Technology, Sri Krishnadevaraya University, AP, India

DOI: 10.5281/zenodo.10054248

ABSTRACT

In FANET's both link failure and node failure can result in packet loss, additionally, the methods for choosing the bypass route and avoiding traffic are rarely used. In order to address congestion and route errors utilizing bypass route selection in FANETs, we offer an adaptive reliable, and congestion control routing protocol in this study. Our proposed protocol gives finer results as compared to the distinguished Protocols corresponding to Congestion control. Numerous routes are built. For effective data transfer, the shortest pathways are found among them. Based on link and route capacity and utilization, congestion is identified. When a source node notices congestion on a connection along the path, it splits traffic using a traffic-splitting function while taking the path availability threshold into account. Using the congestion indicator bit, a node alerts its neighbours if it is unable to resolve the congestion. We demonstrate through simulation that the proposed protocol is dependable and increases throughput while reducing overhead and packet losses.

KEYWORDS: Packet loss, Bypass traffic, Congestion, Threshold**1. INTRODUCTION**

Quality of service (QoS) refers to how well a service performs for a user inside a given network. The majority of multimedia programs have stringent Quality of Service criteria that must be met. The primary goal of QoS provisioning is to achieve more predictable network behaviour, where information from the network can be provided better and network resources can be used efficiently. However, the key difficulties in FANET [2, 11] are in delivering QoS solutions and in sustaining end-to-end QoS with end-user mobility.

Finding a path from source to destination isn't enough for QoS routing; the path must also meet the end-to-end QoS criterion in terms of bandwidth or latency. Maximizing the efficiency with which network resources are used involves calculating routes that are optimal for the various applications that produce network traffic. The primary goals of quality-of-service routing are as follows [3, 12]

- The goal is to locate a route between the two points that meets the needs of the users.
- To get the most out of the internet connection.
- To lighten the stress on the network if abnormal conditions, such as congestion or broken connections, are present.

The primary challenge of FANET is optimizing data packet delivery despite dynamic network architecture without introducing excessive routing complexity. When data packets are routed to obsolete or incorrect pathways due to a broken connection, the packet delivery ratio drops.

In Flying Ad hoc networks, packet loss often occurs due to a failed connection or a node. If there is more than one packet using this link when the physical layer fails, the MAC layer will fail. Even if such packets are removed from the queue after a failure by the routing protocol, fresh packets continue to be added to the queue. If the new packets are sent toward the failing link, they will block all other packets, leading to low throughput and significant delays across the network [5, 7, 16]. Without a command-and-control hub, it's difficult to see

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problems like these in an ad hoc network. Because of this, the malfunctioning nodes stick around for a long period, which slows down the ad hoc network's routing. For example, if a faulty node in the routing process misses data packets, a sizable percentage of packets will be lost [6, 13].

When the volume of network traffic, measured in the form of messages, becomes too great, a condition known as congestion sets in. Wires are what make routing possible in processed by routers at the network's core. Incoming packets are temporarily stored in the routers' buffers before being sent to their final destinations. As a result, multi-port routers might experience congestion and begin losing packets when their buffers fill up. This issue occurs when a router has more incoming traffic than it can process and forward [4].

The CSMA/CA channel access mechanism is used by nodes in wireless networks, and a wireless device serving as a router in a FANET can accept traffic from just one neighbouring device at a time. Due to the busy medium, a device may receive numerous packets before it is able to send a single packet; nonetheless, this is not intended to cause a buffering issue at the device [10].

The primary issue with mobile wireless ad hoc networks is congestion. Congestion is a typical problem in MANET because of the nature of wireless networks and the restricted resources they have access to. Packet transmissions are degraded in such networks owing to interference and fading caused by the shared wireless channel and the dynamic topology. Real-time traffic in big volumes might cause congestion when available bandwidth is depleted. Packet loss, decreased bandwidth, and extra time and power consumption are all results of congestion. The impact of congestion can be mitigated, but the issue cannot be solved entirely. Congestion can be mitigated using appropriate traffic flow processes and rules.

The Internet's TCP congestion management is quite effective. Some of the peculiarities of MANETs have a significant impact on the strategies used to manage congestion. The typical TCP protocol has significant difficulties in a MANET due to the wildly varying conditions. When Internet congestion develops, it usually affects just one router. Congestion, on the other hand, has far-reaching consequences in FANETs because of the shared medium. Congestion prevents any single sender from intentionally bringing down the network in a FANET because of their relatively limited bandwidth. The impact of even a single traffic flow on the network condition can have a profound impact on the fairness of the system as a whole. Wireless multi-hop networks are more vulnerable to overload-related issues than traditional wired networks like the Internet. For this reason, a reliable congestion management protocol is necessary to maintain a satisfactory level of network performance and reliability [14, 15].

In order to identify route failure, a cross-layer proposal called explicit link failure notification (ELFN) involves communication between TCP and the routing protocol. The node that detects the failure sends an ELFN message back to the TCP sender over the routing protocol. TCP sequence numbers, IP addresses, and port numbers are all included in EFLN messages. Therefore, this TCP can tell the difference between congestion-related and mobility-related losses. With the fixed retransmission timeout protocol, route failure is signalled after two consecutive retransmissions due to timeout, and routing error recovery is included. Therefore, the TCP sender will periodically retransmit. To identify network partition and congestion, ad hoc TCP (ATCP) uses the ICMP protocol and the explicit congestion notification (ECN) mechanism. ATCP places TCP into "persistent mode" and retransmits the lost packet from the buffer if it detects three duplicate ACKs. This prevents the TCP congestion control from being activated if it is not necessary. Upon receiving an ECN message, ATCP simply passes the packet along to TCP, which then uses its standard congestion management method [8].

2. Related Work

Without modifying the queuing, scheduling, or forwarding rules of current networks, Peng *et al.* [18] have provided a multi-rate multicast congestion management system for FANETs to achieve excellent fairness with TCP, resilience against misbehaving receivers, and traffic stability. By collecting data and regulating speeds in real-time, this approach only adds a little amount of control traffic overhead.

Supporting congestion management in ad hoc networks with scalable and distributed cluster-based techniques is the goal of the cluster-based congestion control protocol developed by Karunakaran *et al.* [9].

Their strategy relies on the network naturally forming into clusters. Within their limited sphere of influence, clusters may independently and proactively monitor the congestion. By sending a moderate number of control packets, nodes in a cluster may adjust their rates and work together. The clustering method has been applied to deduce the interflow communications. When compared to end-to-end methods, theirs can boost the system's responsiveness.

An explicit rate-based congestion control approach was proposed by Rahman *et al.* [20]. They claim that by enabling the routers to provide explicit feedback, throughput may be quickly enhanced. The sender's flow may be managed with the use of the explicit information included in the feedback packets from the routers. Within the limited scope of FANETs, its protocol outperforms the more cautious TCP and TCP-like protocols when it comes to streaming multimedia. By sending a moderate number of control packets, nodes in a cluster may adjust their rates and work together. The clustering method has been applied to deduce the interflow communications. When compared to end-to-end methods, theirs can boost the system's responsiveness.

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In order to manage FANET congestion, Nishimura *et al.* [17] have suggested a routing protocol that makes use of many agents. They are using two different sorts of agents for the routing process. Their agents do things like collect data on network congestion and connection failure through one of their agents called a routing agent (RA). Message agents (MA) are another type of agent, and they relay the data collected by RA to their intended recipients. With the use of an evaluation function, MAs can determine which way the data packets are travelling.

A wireless greedy primal-dual (wGPD) method for concurrent congestion control and scheduling in FANET has been suggested by Akyol *et al.* [1]. The scheduling choices in wGPD are made in two stages: the first is at the node level, and the second is between nodes. When using intra-node scheduling, every node makes its own decision on which packet to send when its turn comes. Interfering transmissions compete for the right to send the next packet in inter-node scheduling. They have separated congestion control into two categories: reliable and unreliable. The first variant is well suited for the conventional UDP protocol and is employed by loss-tolerant flows. The second variant is suitable for the TCP protocol and guarantees consistent data delivery. Based on our review of current systems, we can say that efforts to alleviate traffic on a by-pass route are uncommon.

3. PROPOSED PROTOCOL: EFCCCRP

The Protocol-Efficiency-based flexible credence congestion control Routing Protocol (EFCCCRP) is a routing protocol that prioritizes efficiency and congestion control in network communications. It is designed to optimize the utilization of network resources, minimize congestion, and ensure smooth data transmission. This protocol incorporates several key elements and features to achieve its objectives.

One of the key characteristics of EFCCCRP is its focus on protocol efficiency. It aims to maximize the utilization of network resources by efficiently routing data packets through the most optimal paths. By minimizing unnecessary overhead and optimizing routing decisions, EFCCCRP enhances the overall efficiency of the network.

Flexibility is another important aspect of EFCCCRP. It is designed to adapt to dynamic network conditions, such as varying traffic loads, changing network topology, and fluctuations in link quality. This adaptability enables EFCCCRP to effectively handle changes in the network and maintain optimal performance.

Congestion control is a crucial element of EFCCCRP. It incorporates mechanisms to detect and control congestion within the network. By monitoring packet delays, buffer utilization, or link utilization, EFCCCRP

can identify potential congestion points and take proactive measures to regulate data flow and prevent congestion-related issues like packet loss and excessive delays.

The protocol also supports Quality of Service (QoS) parameters. This means that different traffic flows can be prioritized based on their specific requirements. By providing QoS support, EFCCCRP ensures that critical or time-sensitive data receive higher priority, resulting in improved overall network performance and user experience.

Scalability is a key consideration in the design of EFCCCRP. It is built to handle network scalability, accommodating a growing number of nodes and network size. This scalability feature ensures that the protocol can effectively scale with the network's expansion without compromising its efficiency and performance.

While EFCCCRP offers several advantages, it's important to be aware of potential limitations and disadvantages. Developing and implementing a new routing protocol can be complex and may require significant engineering effort. Integrating EFCCCRP into existing network infrastructures could pose compatibility challenges, requiring modifications or upgrades to routers and network devices. Additionally, gaining adoption and acceptance of a new routing protocol can be challenging, as it may require convincing network operators and administrators of its benefits over existing protocols.

Overall, EFCCCRP is a Protocol-efficiency-based flexible credence congestion control Routing Protocol that prioritizes efficiency, congestion control, and adaptability. By optimizing resource utilization, adapting to dynamic network conditions, and effectively managing congestion, EFCCCRP aims to enhance the performance and reliability of network communications.

4. COMBINED LINK STABILITY ESTIMATION EMPLOYED BY ROUTING PROTOCOL

4.1. The Signal Strength under Measuring

In a cross-layer architecture, the received signal intensity is determined at the physical layer and made available at higher levels. Reassigning the received signal strength measurement to the MAC layer requires modifying the processes at the physical layers. This information is also utilized in the routing and neighbour tables to make decisions.

The intensity of the signal received is used as an interlayer interaction parameter and sent to the higher layers. The network's performance is enhanced by using the received signal strength to fine-tune the medium access and routing algorithms in accordance with the necessary cross-layer architecture.

The IEEE 802.11 MAC protocol may be trusted. Assuming a maximum transmission power, the received signal intensity must be sufficient to reach all exposed nodes. An RTS (Ready to Send) packet is transmitted with an attached transmission power level by the sender node. While the RTS packet is being received, the receiving node takes a reading of the received signal strength according to the free-space propagation model.

$$P_R = P_T (\lambda / 4\pi d)^2 G_T G_R$$

Where

λ is the wavelength of the carrier,

"d" = distance between sender and receiver,

P_R = Receiving Power

P_T = Transmitting Power

G_T = Unity Gain of Transmitting Omni Directional Antenna

G_R = Unity Gain of Receiving Omni Directional Antenna

4.2. Route expiration time (RET) or Date of Expiry of Route

From among all possible Link Expiry Times (LETs) or Time-Expired Links (or TELs) created for the viable path, the shortest one is chosen as the RET. Connectivity between two nodes is measured in terms of their LET

or TEL [19]. As a result, the routing path with the highest RET count is chosen, which indicates the way is the most dependable.

$$RET_i = \text{Min} (LETs_i)$$

That's why the RET is the cheapest viable LET.

The idea behind LET is to use the movements of two neighbors to get an estimate of when they will become disconnected in the future. The next procedure will show you how to do this. Two neighbouring nodes' motion characteristics can be calculated using a GPS system. All nodes have their clocks synchronized to the GPS reference time, and it is assumed that the signal intensity merely relies on the distance to the transmitter in free space. Using the characteristics of their motion, we may determine how long the connection between the two nodes lasted. Speed, bearing, and signal strength are only a few of the GPS-derived data.

The availability of a link, $L(T_p)$, between two nodes at time t_0 is predicted by the amount of time, $L(T_p)$, that the link will be active. $L(T_p)$ is indicated as

$$L(T_p) = P\{\text{to say valid to } t_0 + T_p | \text{Available at } t_0\}$$

expressing the likelihood that the connection will be up and running from t_0 to $t_0 + T_p$

Since T_p is a reliable forecast, the chance that the epochs from t_0 forward for the two nodes are longer than T_p may be easily computed as $L_1(T_p)$. If the two nodes' motions do not change. Because of the 'memory less' nature of the exponential distribution and the fact that nodes' motions are not reliant on one another, The formula for $L_1(T_p)$ is

$$L(T_p) = L_1(T_p) + L_2(T_p)$$

$$L_1(T_p) = [1 - E(T_p)]^2 = e^{-2\lambda T_p}$$

It is challenging to learn about the changes in link status that result from changes in a node's mobility, which makes it hard to provide an accurate computation for $L_2(T_p)$.

4.3 Node's remaining energy

All nodes are supposed to have a residual power-detecting device and to be aware of their precise physical location. A packet's kinetic energy during transmission may be calculated as

$$Energy_{tx} = P_{size} * Power_{tx} / LBW$$

Where p_{size} is the size of the data packet, $Power_{tx}$ is the power used to send the packet, and LBW is the bandwidth available via the wireless connection? The amount of energy expended in sending a single packet is proportional to the node's distance when it engages in power management during transmission.

$$Energy_{tx} = kd^\alpha$$

Where k is a proportionality constant, d is the separation between two adjacent nodes, and α is an environment-specific parameter (often in the range of 2 and 4). In terms of determining the quantity of energy needed, the shorter the distance between the transmitter and the receiver, the better. The total amount of energy needed is calculated at each node as

$$Energy_{tot} = p * (Energy_{tx} + Energy_{pro})$$

Depending on how many packets there is (p). When compared to packet transmission, packet processing has a significantly lower energy need ($Energy_{proc}$). The energy that remains in a node following packet transmission is known as its residual energy or remaining energy. Power comes from

$$Energy_{res} = Energy_{initial} - Energy_{tot}$$

4.4 Combined metric:

Finally, the combined metric for finding the link lifetime is given by

$$CRM = P_R + L_1(T_p) + Energy_{res} + LA_N$$

The node status is adaptively determined based on the Value of CRM as given below.

Node status is Green if $CRM_{min} < CRM < CRM_{max}$

Node status is yellow if $CRM = CRM_{min}$

Node status is red, if $CRM < CRM_{min}$

Where CRM_{min} and CRM_{max} are the maximum and minimum combined routing metric values.

Then, we carry out Bypass Route Discovery as we did in our earlier work. In the event of a link failure, the upstream node with the cached data in its buffer can retransmit the packets via a bypass route, which takes the data from the next reliable connection instead of the failed one.

5. CONGESTION MITIGATION THROUGH PROPOSED BYPASS ROUTE

5.1 Multipath construction

The network may be thought of as a directed graph $G = (V, E)$, where V is the set of nodes and E is the set of links or edges. Our method proposes locating various routes by searching for shortest and disjoint routes.

5.1.1 Fastest routes

The algorithm for finding the shortest path between two points, S and D , is called the shortest path discovery algorithm. Let's say the shortest pathways between S and D are denoted by SP_{sd}

The collection of nodes next to node S is denoted by N_S . Then, we may calculate approximations for the alternative shortest pathways with the formula

$$s^{P_{sd}^{s-1}} = \{S\} \oplus S_{s-1}^P, d$$

Concatenation of two finite sub-paths, where $s - 1$ is the neighbor node of S . Starting at node S , the shortest route S_{s-1}^P, d leads to node D through the neighbour node $s - 1$. This method of discovering new routes through nearby nodes might, however, introduce the routing loop issue.

As such, S_{sd}^P must ensure that it does not include S as an intermediary node for the generated shortest routes to be loop-free. Therefore, the requirement stated in Algorithm-1 must be satisfied by any shortest path.

Algorithm-1

Assumption: S and D as the source and destination nodes, Respectively

S_{sd}^P be the set of shortest paths between nodes S and D

1. if $(S \notin \text{nodes}(S_{s-1}^P, d))$ then
2. The alternate path $s^{P_{sd}^{s-1}}$ will be loop free
3. else
4. The alternate path $s^{P_{sd}^{s-1}}$ sd will be routing loop path
5. end if

A_{sd} contains all the calculated trajectories that branch of nodes S and D . Included paths, or those belonging to the set A_{sd} , can be written as follows:

$$A_{s,d} = \{ s^{P_{sd}^{s-1}} : s-1 \in N_s, S \notin s^{P_{sd}^{s-1}} \}$$

Therefore, the set A_{sd} has several possible alternatives for getting from S to D.

5.1.2 Disjoint paths

Multipath routing must meet the disjoint pathways criterion for it to be effective and reliable. Any pair of pathways in A_{sd} is disjoint if and only if the following criteria hold.

Algorithm-2

Assumption: Let S and D denotes the source and destination nodes respectively

Let $s^{P_{sd}^{s-1}}$ and $s^{P_{sd}^{s-2}}$ be two paths in the alternate path set A_s , d

1. Find common nodes in $s^{P_{sd}^{s-1}}$ and $s^{P_{sd}^{s-2}}$
2. If the only common nodes are S and D then
3. The paths will be disjoint paths
4. Else if it has common nodes other than S and D then
5. The paths will be interconnected paths
6. End if

Thus, disjoint paths between S and D can be given as,

$$D_{s,d} = \{sP_{s,d}\} \cup A_{s,d}$$

5.2 Conjoint-c paths

Using this method, a set $cP_{s,d}$ is defined to produce a number of possible pathways. The pathways with the most conjoint linkages are included. Let P_1 and P_2 be two pathways; if the set $cP_{s,d}$ meets the following condition, then P_1 and P_2 can be added to it.

$$|l(P_1) \cap l(P_2)| \leq c$$

Where 'l' denotes linkages and c stands for connections that are shared.

The set $cP_{s,d}$ can be defined in several ways. Paths connecting S and D that are not shared by any other path in the conjoint set $cP_{s,d}$, which exists at Conjoint level 0. Paths with at least one common ancestor are included in $cP_{s,d}$ at Conjoint Level 1. Paths with c common links make up the $cP_{s,d}$ set at Conjoint level-c. A complete definition of the conjoint property is then given as

$$cP_{s,d}^{c=0} \subseteq cP_{s,d}^{c=1} \subseteq \dots \subseteq cP_{s,d}^{c=c} \subseteq cP_{s,d}$$

Below, in **Algorithm-3**, is the algorithm I used to generate the $cP_{s,d}$ set.

1. Initialize $cP_{s,d}$ to $sP_{s,d}$
2. paths in $A_{s,d}$ are arranged in ascending order of their path length
3. further paths for the set $cP_{s,d}$ is selected from the set $A_{s,d}$
4. while ($A_{s,d} \neq 0$) do
5. find out the shortest path sp in $A_{s,d}$
6. Remove sp from $A_{s,d}$
7. Break
8. Check path sp against all paths in $A_{s,d}$
9. Check whether it satisfies conjoint condition (equation-15)
10. If sp satisfies condition (equation-15) for all paths in $cP_{s,d}$ then
11. Add sp to the set $cP_{s,d}$
12. End

5.3. Load Balancing

We provide a solution to the network congestion using Algorithm 1, which involves sending data packets over group-G multipath paths. When congestion is detected on a local outbound link L_n for which the path p_{ND} contains link L_n , node N determines the multipath routes to destination D.

Then, $P \in P^G$ some of the data destined for node D is rerouted onto other networks.

Let's pretend that each node in the network (let's call it node N) keeps track of how often it uses its own outbound connection, L_n . Path utilization is represented by the sign P_{Ut} , and the link utilization is represented by the symbol L_{Ut} . The percentage of time spent on route p1 may be calculated as

$$P_{Ut}(p_1) = \min \{L_{Ut}(L_n); L_n \in P^G\}$$

Each node accounts for network or routing information relayed by its neighbors while determining its own path and link utilisations.

$$P_{Ut}(p_{s,d}) = \min \{L_{Ut}(L_{s,i}), P_{Ut}(p_{j,d})\}$$

Congestion is identified at the link level if the use of that link rises over a local congestion threshold T_H . In this case, T_H represents a fixed threshold. The primary objective is to lower utilization enough by rerouting some of the traffic via alternate routes (also known as bypass traffic). When a node detects local link congestion or gets an Absolute Congestion Index (ACI) bit from a neighbor, it computes a list of alternate pathways and distributes the bypass traffic among these paths. When a node is unable to alleviate congestion using the first technique, it sends signals to its neighbors using the ACI bit. To alleviate bottlenecks on the local link, congestion-triggered multipath traffic dispersal employs the steps outlined below.

First, if node N sees that the outgoing link l is congested locally ($l, P_{Ut}(L_n) > T_H$), or if it gets an ACI = 1 bit from a neighbor node on link l, it will attempt to reroute traffic onto other pathways to avoid the congested link. Group-G pathways found using Algorithm 1 are the alternatives.

Secondly, we define $SP = \{p \in P^G : P_{Ut}(p) < \mu\}$ where μ is the path availability threshold.

Third, split the data transfer across the SP paths.

The traffic on node N's active path P_a is distributed evenly among all of its accessible paths. Assume that node N is now experiencing 100 kbit/s of demand. Let's pretend SP has four possible routes labelled P_1, P_2, P_3 , and P_4 . Upon receiving an Explicit Congestion Indication (ECI), node N will start sending data along other pathways in addition to its principal path P_a , at a rate of $100/4 = 25$ kbit/s.

5.4 Pseudocode and Flowchart for the EFCCCRP (Combined Link Stability & Congestion control)

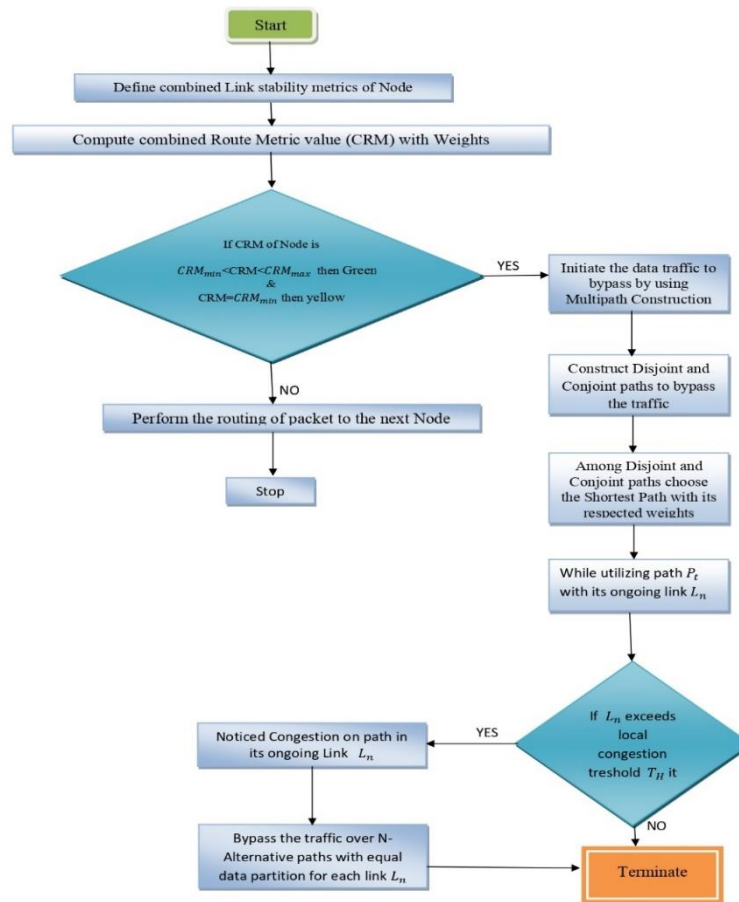


Figure 1: A Schematic Representation of Link Stability and Congestion Control in EECCRP

Begin

Link Stability:

- Define the metrics values of the node ($P_R, L_1(T_p), Energy_{res}, LA_N$)
- Compute CRM value $CRM = P_R + L_1(T_p) + Energy_{res} + LA_N$
- If (CRM is $CRM_{min} < CRM < CRM_{max}$)
- Status is Green then
- Utilise the path for data transmission
- Else if (CRM is $CRM = CRM_{min}$)
- Status is yellow then there is a slight possibility of congestion so
- Wait random amount of time t_0 then choose suitable path for data transmission

Multipath Construction:

- Link is likely to get broken if it is red then bypass the traffic through multipath construction by using conjoint and disjoint paths
- Construct disjoint paths between S and D can be given as, $D_{s,d} = \{sP_{s,d}\} \cup A_{s,d}$
- Construct conjoint paths by using $cP_{s,d}^{c=0} \subseteq cP_{s,d}^{c=1} \subseteq \dots \subseteq cP_{s,d}^{c=c} \subseteq cP_{s,d}$
- Choose the alternative shortest path from both conjoint and disjoint paths
- Shortest path is given by: $A_{s,d} = \{s^{psd-1} : s-1 \in N_s, S \notin s^{psd-1}\}$

- f) While bypassing the traffic over G-Path routes
- g) If the path utilization gets $ACI=1$ or $(l, P_{UT}(Ln) > T_H)$, in ongoing path utilization in link exceeds threshold value
- h) The it indicates congestion has been occurred in the ongoing link
- i) Re-route traffic with $SP = \{p \in P^G : P_{UT}(p) < \mu\}$ where μ is the path availability threshold
- j) Split the data transfer to route with equal partitions across the SP paths.
- k) End

6. SIMULATION RESULTS

6.1 Simulation model and parameters

To test our methodology in a virtual environment, we employ NS2. All of the mobile hosts in our scenario have their channel capacity fixed at 2 Mbps. As the media access control (MAC) layer protocol, we implement IEEE 802.11's distributed coordination function (DCF) for wireless LANs. It can signal the network layer when a connection fails.

We maintain 150 moving nodes in our simulation. In a simulation lasting 50 seconds, the mobile nodes cover an area measuring 1000 by 9000 metres. Let's pretend every node in the network travels at the same pace on average. The 250-meter transmission range is standard across all nodes. Our model projects a speed of 10 metres per second. The simulated traffic is CBR, which stands for Constant Bit Rate. The number of CBR flows might be 2, 4, 6, or 8.

6.2 Measures of Performance

The following measures are used most often in our evaluations of performance.

- Time lag between the beginning and end of all data packets that make it from their sources to their destinations are averaged to get the total round-trip latency.
- Rate of packet delivery on average It's the ratio of successful packet receptions to total packet transmissions.
- Drop is the typical amount of lost data packets.
- Speed of Throughput The term "throughput" is used to describe the rate at which routing control packets are successfully received.
- Overhead It is the total number of control packets sent and received during a data packet transmission.

In the next part, the outcomes of the simulation will be shown.

The IS-IS method is also implemented in NS-2 with the same simulation parameters. Next, we evaluate the IS-IS method and the established AODV protocol in contrast to the suggested adaptive reliable congestion control routing protocol (EFCCCRP).

6. RESULTS ANALYSIS:

The number of nodes in the network is increased from 50 to 100 to 150 to 200 to 250 and 300 to 350 to 400 to 450 to 500 in this simulation. We maintain a constant 8 flows and 250 kb/s data rate.

Adding more nodes results in a linear increase in latency, as seen in Figure. 2. The most noticeable delay is caused by AODV, followed by IS-IS. When compared to IS-IS and AODV, EFCCCRP has a substantially shorter latency.

The packet delivery ratio drops significantly as the number of nodes grows larger because more packets are lost. As can be seen in Figure. 3, EFCCCRP has the best delivery ratio, followed by IS-IS, while AODV has the lowest delivery ratio. Figure 4 demonstrates that EFCCCRP and AODV have the lowest packet loss, whereas AODV has the greatest.

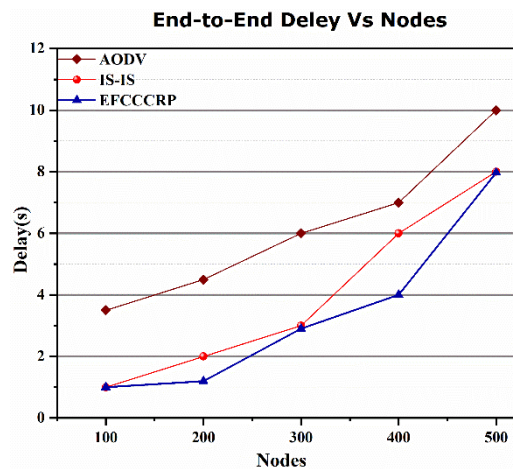


Figure 2: End-to-End Delay

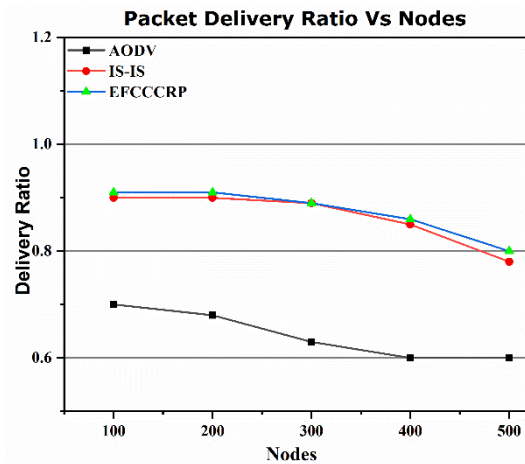


Figure 3: Packet Delivery Ratio

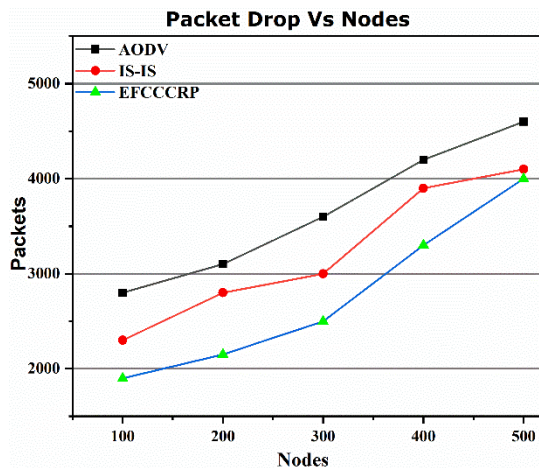


Figure 4: Packet Drop

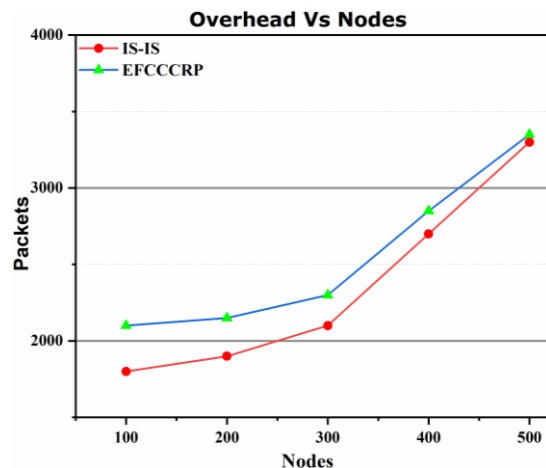


Figure 5: Overhead

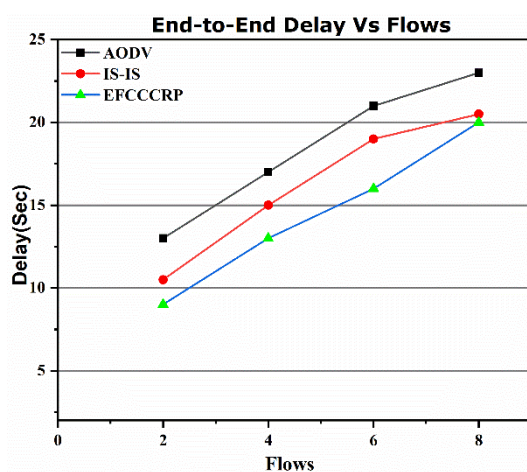


Figure 6: End-to-End Delay Vs Flows

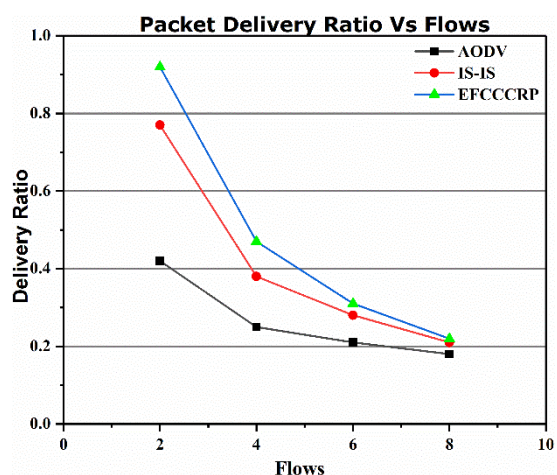


Figure 7: Packet Delivery Ratio Vs Flows

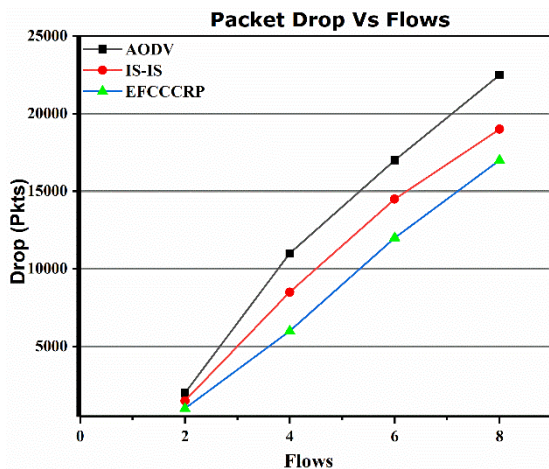


Figure 8: Packet Drop Vs Flows

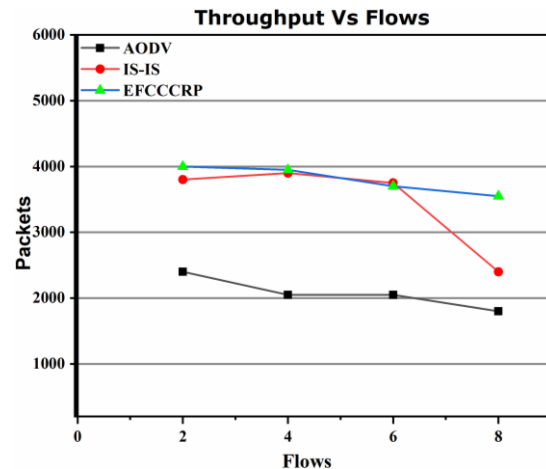


Figure 9: Throughput Vs Flows

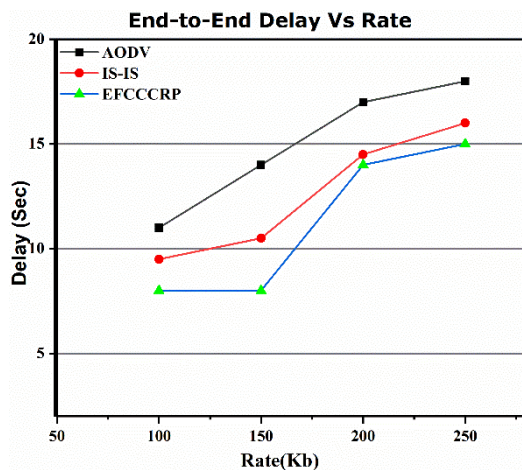


Figure 10: End-to-End Delay Vs Rate

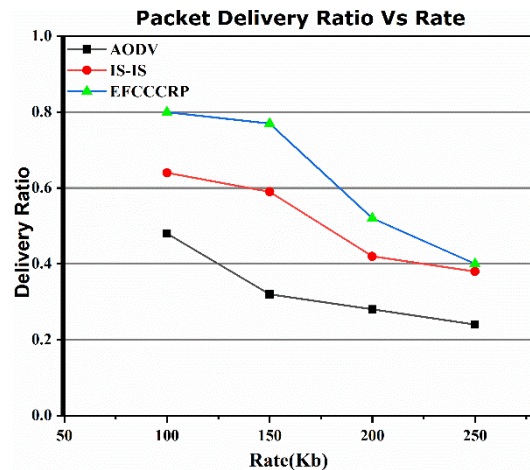


Figure 11: Packet Delivery Ratio Vs Rate

Overhead grows proportionally with the number of nodes since more nodes mean more control packets need to be sent. However, as can be shown in Figure. 5, EFCCCRP has a fraction of the overhead of the standard IS-IS protocol.

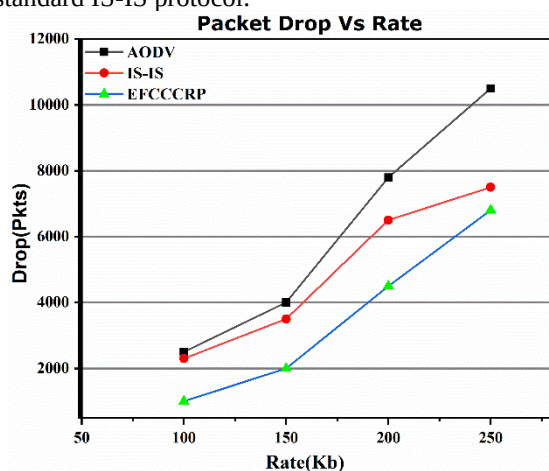


Figure 12: Packet Drop Vs Rate

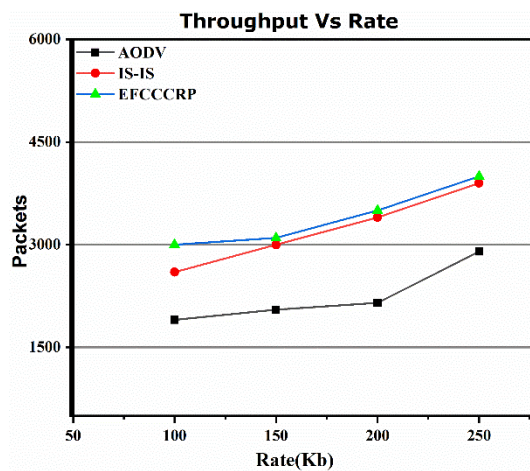


Figure 13: Throughput Vs Rate

The end-to-end delay is proportionally Figure. 6 reduced in EFCCCRP when the number of flows is gradually increasing where as in AODV and IS-IS the delay is increasing based on the increased number of flows.

Table 1. Delay versus Nodes

AODV		IS-IS	EFCCCRP
Node s	Delay(s)	Delay(s)	Delay(s)
100	3.5	1	1
200	4.5	2	1.2
300	6	3	2
400	7	6	3
500	10	9	7

Table 2. Delivery Ratio versus Nodes

AODV		IS-IS	EFCCCRP
Node s	Delivery Ratio	Delivery Ratio	Delivery Ratio
100	0.7	0.9	0.91
200	0.68	0.9	0.91
300	0.63	0.89	0.95
400	0.6	0.85	0.88
500	0.6	0.78	0.8

Table 3. Packet Drop versus Nodes

AODV		IS-IS	EFCCCRP
Node s	Packet s	Packet s	Packets
100	2800	2300	1900
200	3100	2800	2150
300	3600	3000	2700
400	4200	3900	3700
500	5200	4100	4000

Table 4. Overhead versus Nodes

IS-IS		EFCCCRP
Node s	Packet s	Packets
100	1800	2100
200	1900	2200
300	2100	2300
400	2700	2900
500	3300	3400

Table 5. Delays versus Flows

AODV		IS-IS	EFCCCRP
Flow s	Delay (Sec)	Delay (Sec)	Delay (Sec)
2	13	10.5	4.5
4	17	15	14
6	21	19	16
8	23	20.5	19.5

Table 6. Delivery Ratio versus Flows

AODV		IS-IS	EFCCCRP
Flow s	Delivery Ratio	Delivery Ratio	Delivery Ratio
2	0.42	0.77	0.92
4	0.25	0.38	0.47
6	0.21	0.28	0.31
8	0.18	0.21	0.22

Table 7. Packet Drop versus Flows

AODV		IS-IS	EFCCCRP
Flow s	Drop Packet	Drop Packets	Drop Packets
2	2000	1500	1000
4	1100	8500	6000
6	1700	14500	12000
8	2250	19000	17000

Table 8. Throughput versus Flows

AODV		IS-IS	EFCCCRP
Flow s	Packet s	Packet s	Packets
2	2400	3800	4500
4	2050	3900	4800
6	2050	3750	4400
8	1800	2400	3250

Table 9. Delay versus Rate

AODV		IS-IS	EFCCCRP
Rate(Kb)	Delay(Sec)	Delay(Sec)	Delay (Sec)
100	11	9.5	4
150	14	10.5	8
200	17	14.5	12.5
250	18	16	15

The packet delivery ratio Figure.7 is also proportionally increased in EFCCCRP when the number of flows is gradually increasing when compared with AODV and IS-IS. The packet drops are gradually decreased Figure.8 based upon the number of flows in EFCCCRP when compared with AODV and IS-IS and also when the number of packets is increasing Figure.9 along with the number of flows the throughput is gradually Figure.10 increased when compared with the standard protocols AODV and IS-IS.

Table 10 Delivery Ratio versus Rate

AODV		IS-IS	EFCCCRP
Rate (Kb)	Delivery Ratio	Delivery Ratio	Delivery Ratio
100	0.48	0.64	0.8
150	0.32	0.59	0.77
200	0.28	0.42	0.52
250	0.24	0.38	0.4

Table 11 Packet Drop versus Rate

AODV		IS-IS	EFCCCRP
Rate (Kb)	Drop (Pkts)	Drop (Pkts)	Drop (Pkts)
100	2500	2300	1000
150	4000	3500	2000
200	7800	6500	4500
250	10500	7500	6800

Table 12 Throughput versus Rate

AODV		IS-IS	EFCCCRP
Rate (Kb)	Packet s	Packet s	Packets
100	1900	2600	3000
150	2050	3000	3200
200	2150	3400	3730
250	2900	3900	4100

The delay in transmission of packets at the Figure.10 rate of 250kb is very low in EFCCCRP when compared with AODV and IS-IS and in view of the metrics like packet delivery ratio, Figure.11 packet drop, throughput when those metrics are compared Figure.12 with the data rate of transmission that is done from source to destination EFCCCRP Figure.13 gives the better results and those metrics performance are better when compared with AODV and IS-IS.

7.1 Comparing the EFCCCRP with Other Standard Protocols

When comparing EFCCCRP with AODV, and IS-IS from the perspective of congestion control, EFCCCRP stands out with its explicit focus on congestion control mechanisms. EFCCCRP incorporates dedicated congestion control techniques, such as credence-based congestion control, to regulate data flow and prevent congestion within the network. It employs adaptive mechanisms to detect and respond to congestion in real-time, optimizing the utilization of network resources and ensuring smooth data transmission.

On the other hand, EFCCCRP, AODV, and IS-IS do not have explicit congestion control mechanisms. EFCCCRP relies primarily on reactive measures, such as route rediscovery to handle congestion situations. AODV lacks built-in congestion control mechanisms, which can potentially lead to congestion-related issues. Similarly, IS-IS does not have explicit congestion control mechanisms and depends on other mechanisms for congestion avoidance.

In terms of adaptability, EFCCCRP has more flexibility compared to IS-IS and AODV. EFCCCRP is designed to adapt to changes in network conditions, such as varying traffic loads and link quality, enabling it to effectively manage congestion in dynamic environments.

Additionally, EFCCCRP supports Quality of Service (QoS) parameters, allowing for traffic differentiation and prioritization. This QoS support aids in congestion control by ensuring critical or time-sensitive data receives higher priority, minimizing the risk of congestion-related issues. In contrast, EFCCCRP, AODV, and IS-IS do not natively support QoS differentiation.

Overall, EFCCCRP distinguishes itself by incorporating congestion control mechanisms, adaptability, and QoS support. While EFCCCRP, AODV, and IS-IS have their own strengths, EFCCCRP's explicit focus on congestion control makes it a favourable choice for managing congestion and ensuring efficient data flow within the network. The comparison table based on the specified parameters, highlighting the strengths of EFCCCRP.

7.2 Strengths of EFCCCRP:

Congestion Control: EFCCCRP's explicit focus on congestion control mechanisms ensures optimized throughput and higher packet delivery ratios compared to AODV and IS-IS.

End-to-End Delay: The adaptive mechanisms in EFCCCRP help minimize end-to-end delay during congestion, outperforming AODV and IS-IS in this aspect.

Packet Drop: EFCCCRP's dedicated measures to prevent packet drops give it an advantage over AODV and IS-IS, which may experience increased drops during congestion.

Data Rate: EFCCCRP's optimal resource utilization results in a more consistent data rate compared to AODV and IS-IS, where congestion-related data rate reductions can occur.

Table 13. Comparison Table

Metric	EFCCCRP	AODV	IS-IS
<i>Throughput</i>	Explicit congestion control for optimized throughput	Limited throughput optimization during congestion	Might not optimize throughput during congestion
<i>Packet Delivery Ratio</i>	Real-time congestion control for a higher delivery ratio	Lower delivery ratio possible due to congestion	Congestion might impact the delivery ratio

<i>Overhead</i>	Efficient control mechanisms for reduced overhead	Control overhead might increase during congestion	Overhead might increase during congestion
<i>End-to-End Delay</i>	Adaptive mechanisms to minimize delay in congestion	Congestion-related issues can increase delay	Might not actively manage end-to-end delay
<i>Packet Drop</i>	Dedicated measures to prevent packet drops	Increased packet drops are possible during congestion	Congestion-related issues might cause packet drops
<i>Data Rate</i>	Optimal resource utilization for consistent data rate	Data rate reduction possible due to congestion	Data rate might be affected during congestion

By focusing on congestion control, adaptability, and QoS support, EFCCCRP demonstrates strengths across these metrics, making it a compelling choice for scenarios where congestion management is crucial for maintaining network performance. However, the choice of the "best" protocol depends on specific network requirements and considerations.

7. CONCLUSION

In this research, we present an improved EFCCCRP protocol for bypassing route selection in FANETs to alleviate congestion. A node determines the multipath routes to destinations for which the path contains link L when it notices congestion on a local outgoing link L. Then, a portion of the traffic to the node is diverted to alternate routes. If a local link's utilization rises above the local congestion threshold T_H , congestion is identified on that link. By moving some traffic to alternate routes and treating this traffic as bypass traffic, the goal is to reduce utilization to a level that is more tolerable. Whenever it senses local connection congestion, a node determines a set of alternate paths and distributes the bypass traffic over these paths.

8. ACKNOWLEDGEMENTS

It has been a privilege and a joy working with Prof. V. Raghunatha Reddy throughout my research journey. I am grateful for his guidance, encouragement, and unwavering support. I have been constantly inspired by his dedication and expertise. As a supervisor, He has also been a role model, consistently demonstrating the highest standards of professionalism and commitment.

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